Appendix A: Bayesian Chronological Models

This is an overview of the four Bayesian chronological models used in Sections 2.3 and 2.4 of the Main Text. All radiocarbon dates used in these models can be found in the .csv file in Appendix A, along with contextual information following the recommendations by Bayliss (2015). The model code stems directly from OxCal v4.4.4 (Bronk Ramsey 2009) and all calibrations have been performed with the IntCal 20 curve (Reimer et al. 2021).

1 Analysis of the Funnel Beaker West typochronology

The Bayesian chronological model below tests the typochronology for Funnel Beaker West ceramics proposed by Brindley (1986) against the available radiocarbon dates in Appendix A which are associated with Funnel Beaker West ceramics from a specific horizon (see Section 2.3 of the Main Text).

Model code

```
Plot()
 {
  Sequence("FBW_typochronology")
  {
   Boundary("Start horizon 4");
   Phase("Horizon 4")
   {
    R_Date("GrN-5103",4355,45);
    R_Date("GrA-39456",4485,30);
    R_Date("GrN-16492",4700,40);
    R_Date("GrN-16494",4680,60);
    R_Date("Poz 71757",4460,40);
   };
   Boundary("Boundary 4/5");
   Phase("Horizon 5")
   {
    R_Date("GrN-1824C",4410,60);
    R_Date("GrN-18785",4460,80);
    R_Date("GrN-12262",4920,70);
    R_Date("GrN-12263",4490,60);
    R_Date("GrN-11763",4980,60);
    R_Date("GrN-11764",4400,60);
    R_Date("GrN-11765",4890,80);
    R_Date("GrN-11766",4480,60);
```

```
  R_Date("GrN-9209",4520,35);
  R_Date("GrM-26730",4387,27);
  };
  Boundary("Boundary 5/6");
  Phase("Horizon 6")
  {
  R_Date("GrA-41638",4260,35);
  R_Date("GrA-41639",4230,35);
  R_Date("GrA-13600",4160,50);
  R_Date("GrN-6156",4380,40);
  };
  Boundary("Boundary 6/7");
  Phase("Horizon 7")
  {
  R_Date("GrN-4201",4380,75);
  R_Date("GrN-5767",4315,60);
  R_Date("Poz-88040",4205,35);
  R_Date("GrA-13599",4130,50);
  R_Date("GrA-13705",4200,50);
  R_Date("GrA-13958",4220,50);
  R_Date("GrA-16021",4230,30);
  R_Date("KIA-23204",4348,30);
  R_Date("KIA-23207",4331,29);
  R_Date("R-combine",4150,35);
  R_Date("GrA-13706",4170,50);
  R_Date("GrA-14086",4170,40);
  R_Date("GrA-14088",4270,40);
  R_Date("GrA-14089",4190,35);
  R_Date("GrA-14090",4140,40);
  R_Date("GrA-14095",4235,40);
  R_Date("GrA-14168",4170,40);
  R_Date("GrA-14973",4275,45);
  R_Date("GrA-15642",4390,60);
  R_Date("GrA-15459",4290,60);
  };
  Boundary("End hoirzon 7");
 };
};
```

Model output

The model cannot be resolved in OxCal because the MCMC Sampling process cannot order the dates such that all model constraints are satisfied (see Section 2.3 of the Main Text).

2 Analysis of the Corded Ware typochronology

The current Corded Ware typochronology was formulated by Drenth and Lanting (1991). The Bayesian chronological model below tests this typochronology against all radiocarbon dates associated with Corded Ware and All-Over Ornamented ceramics in Appendix A.

Model code

```
Plot()
 {
  Sequence("CordedWare_Drenth")
  {
   Boundary("Start CW");
   Phase("Early CW")
   {
    Boundary("Boundary phase 1/2");
    Phase("Phase 2")
    {
     R_Date("GrN-6129",4165,55);
     R_Date("GrN-6724",4210,40);
    };
    Boundary("End phase 2");
    R_Date("GrN-1855",4420,55);
   };
   Boundary("Boundary early CW/late CW");
   Phase("Late CW")
   {
   };
   Boundary("Boundary phase 3/4");
   Phase("phase 4")
   {
    R_Date("GrN-11918",4165,30);
    R_Date("GrA-14965",4065,45);
    R_Date("GrN-6368",3935,35);
    R_Date("GrN-6349",3945,40);
    R_Date("GrA-41646",4075,30);
    R_Date("SUERC-26360",4055,35);
    R_Date("GrA-56014",4100,40);
    R_Date("GrN-56013",4030,40);
    R_Date("GrA-12299",3920,60);
   };
   Boundary("End phase 4");
   R_Date("GrN-6126",3970,35);
   R_Date("GrN-8247",4040,90);
   R_Date("UtC-1945",4040,70);
```

```
  R_Date("UtC-1949",3910,50);
  R_Date("GrN-851",4140,70);
  R_Date("GrN-7518",4170,45);
 };
 Boundary("End CW");
};
```

Model output

The model cannot be resolved in OxCal because the MCMC Sampling process cannot order the dates such that all model constraints are satisfied (see Section 2.3 of the Main Text).

3 Overlap between Funnel Beaker West and Corded Ware

The third Bayesian chronological model tests whether overlap exists between Corded Ware and Funnel Beaker West communities on the basis of all available radiocarbon evidence (see Appendix A; Section 2.3 in the Main Text).

Model code

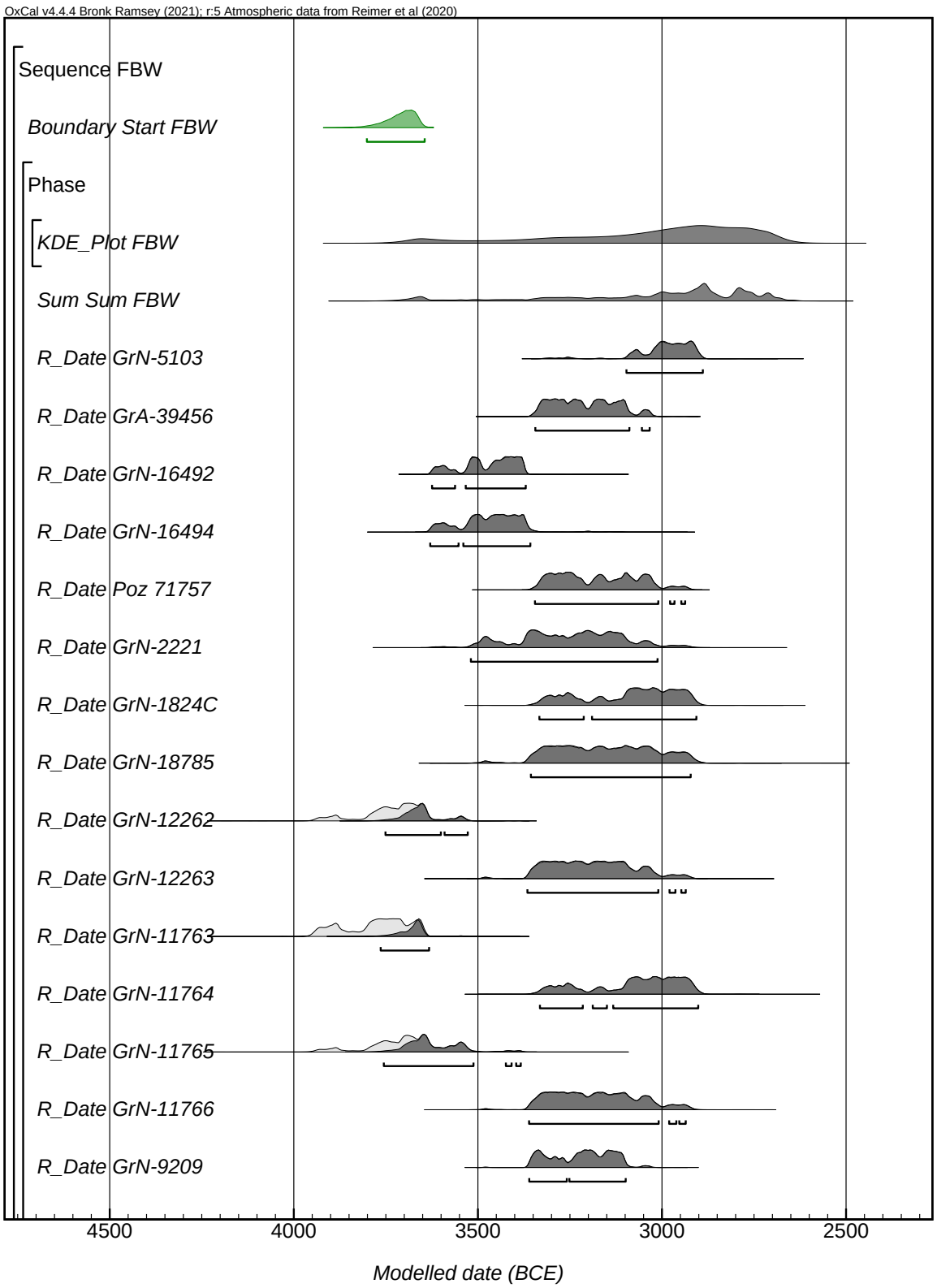
```
Plot()
 {
  Sequence("FBW")
  {
   Boundary("Start FBW")
   {
    color="green";
   };
   Phase()
   {
    KDE_Plot("FBW")
    {
    };
    Sum("Sum FBW")
    {
    };
    R_Date("GrN-5103",4355,45);
    R_Date("GrA-39456",4485,30);
    R_Date("GrN-16492",4700,40);
    R_Date("GrN-16494",4680,60);
    R_Date("Poz 71757",4460,40);
    R_Date("GrN-2221",4550,80);
    R_Date("GrN-1824C",4410,60);
    R_Date("GrN-18785",4460,80);
```

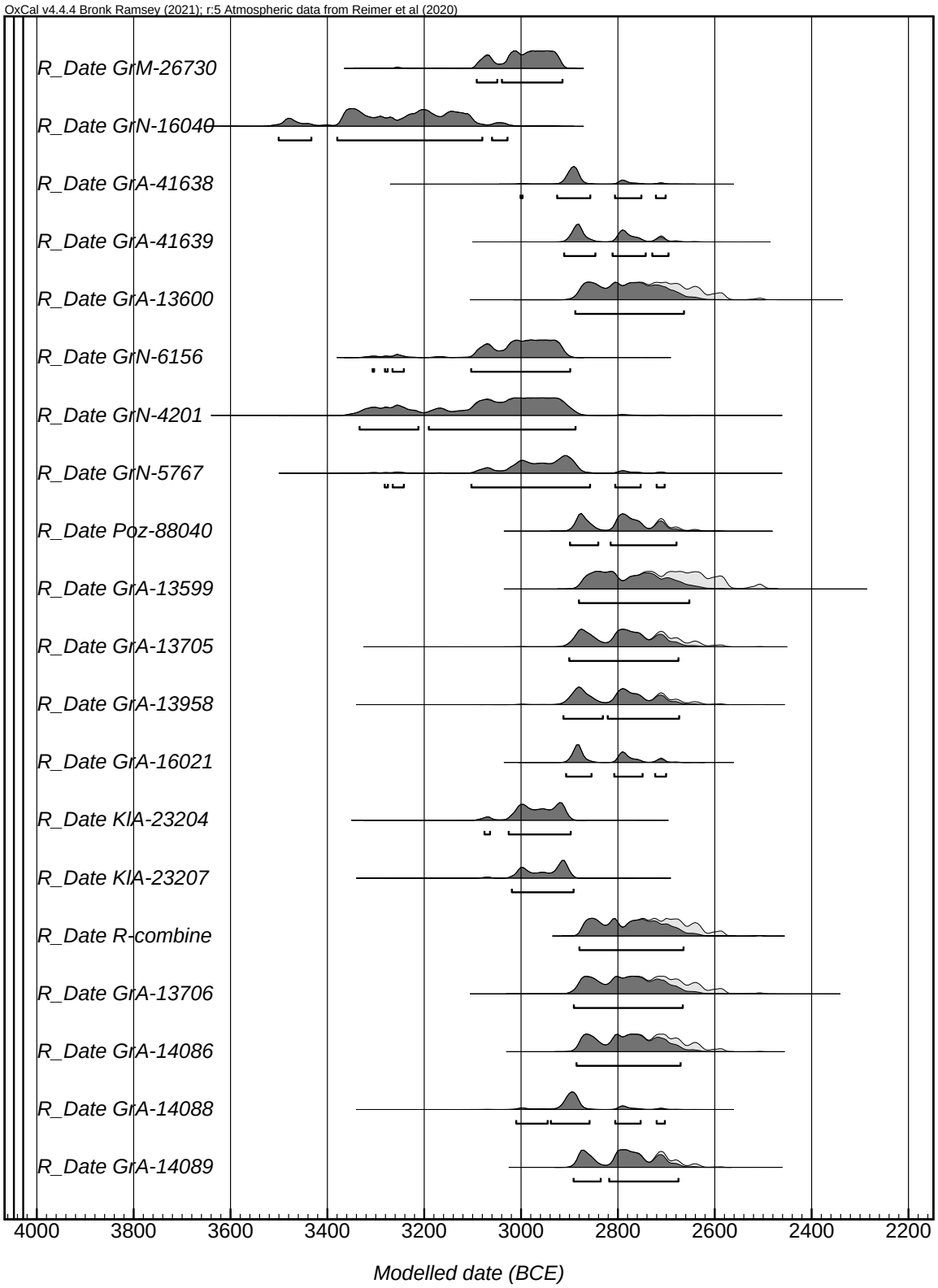
```
 R_Date("GrN-12262",4920,70);
 R_Date("GrN-12263",4490,60);
 R_Date("GrN-11763",4980,60);
 R_Date("GrN-11764",4400,60);
 R_Date("GrN-11765",4890,80);
 R_Date("GrN-11766",4480,60);
 R_Date("GrN-9209",4520,35);
 R_Date("GrM-26730",4387,27);
 R_Date("GrN-16040",4550,60);
 R_Date("GrA-41638",4260,35);
 R_Date("GrA-41639",4230,35);
 R_Date("GrA-13600",4160,50);
 R_Date("GrN-6156",4380,40);
 R_Date("GrN-4201",4380,75);
 R_Date("GrN-5767",4315,60);
 R_Date("Poz-88040",4205,35);
 R_Date("GrA-13599",4130,50);
 R_Date("GrA-13705",4200,50);
 R_Date("GrA-13958",4220,50);
 R_Date("GrA-16021",4230,30);
 R_Date("KlA-23204",4348,30);
 R_Date("KlA-23207",4331,29);
 R_Date("R-combine",4150,35);
 R_Date("GrA-13706",4170,50);
 R_Date("GrA-14086",4170,40);
 R_Date("GrA-14088",4270,40);
 R_Date("GrA-14089",4190,35);
 R_Date("GrA-14090",4140,40);
 R_Date("GrA-14095",4235,40);
 R_Date("GrA-14168",4170,40);
 R_Date("GrA-14973",4275,45);
 R_Date("GrA-15642",4390,60);
 R_Date("GrA-15459",4290,60);
 R_Date("Poz-68248",4330,35);
 R_Date("Poz-68249",4220,35);
 R_Date("Beta 421914",4400,30);
 };
 Boundary("End FBW")
 {
  color="red";
 };
};
Sequence("CW")
{
 Boundary("start CW")
 {
  color="green";
```

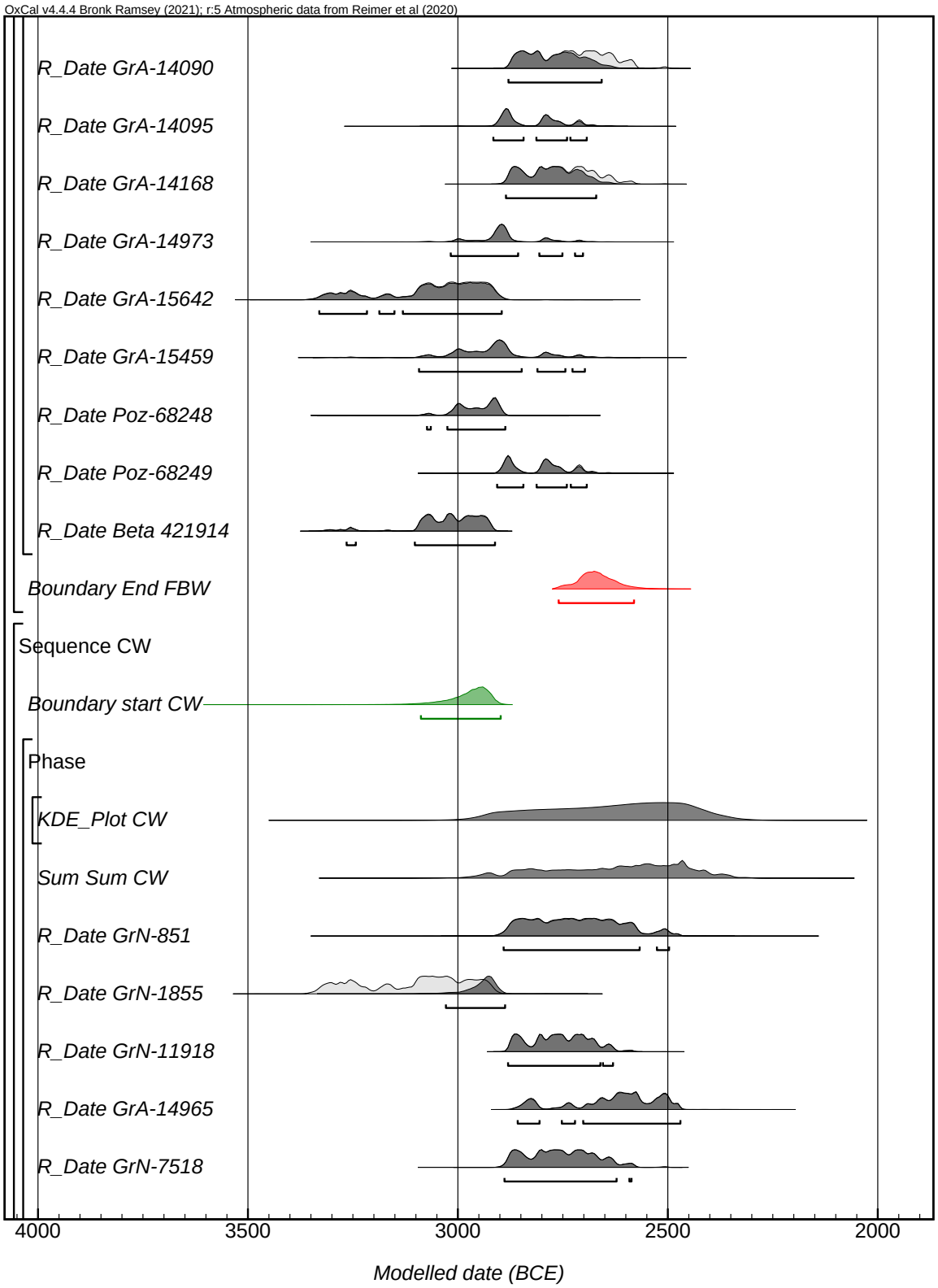
```
 };
 Phase()
 {
  KDE_Plot("CW")
  {
  };
  Sum("Sum CW")
  {
  };
  R_Date("GrN-851",4140,70);
  R_Date("GrN-1855",4420,55);
  R_Date("GrN-11918",4165,30);
  R_Date("GrA-14965",4065,45);
  R_Date("GrN-7518",4170,45);
  R_Date("GrN-6129",4165,55);
  R_Date("GrN-6368",3935,35);
  R_Date("GrN-6349",3945,40);
  R_Date("GrA-41646",4075,30);
  R_Date("SUERC-26360",4055,35);
  R_Date("GrN-6126",3970,35);
  R_Date("GrN-8247",4040,90);
  R_Date("GrN-6724",4210,40);
  R_Date("UtC-1945",4040,70);
  R_Date("UtC-1949",3910,50);
  R_Date("GrA-56014",4100,40);
  R_Date("GrN-56013",4030,40);
  R_Date("GrA-12299",3920,60);
 };
 Boundary("End CW")
 {
  color="red";
 };
};
};
```

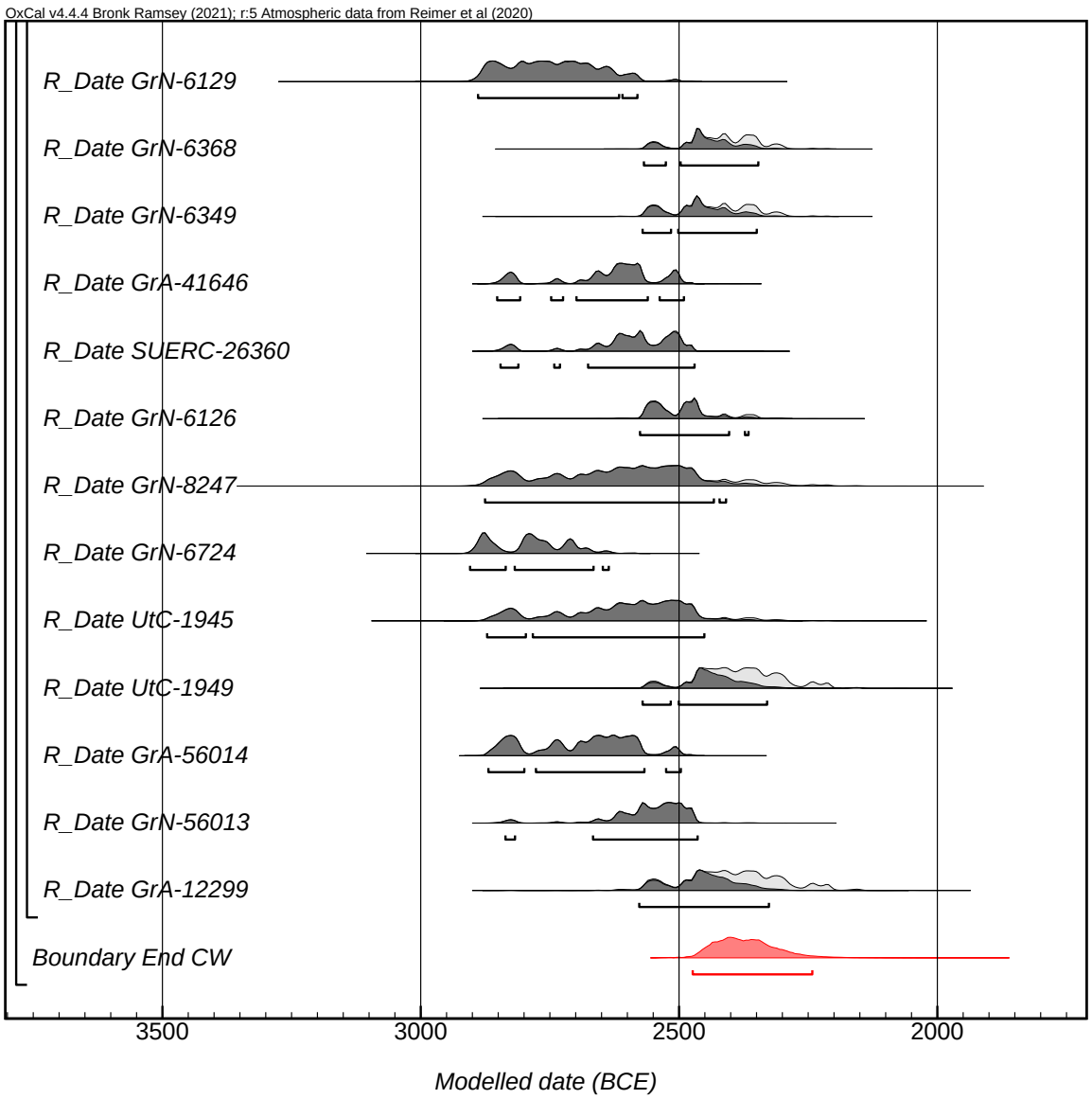
Model output

The outcomes of the analysis have been attached below. The model has good fit: A_{model}=113.1, as do the individual dates $A_{overall}$=111.6. Figure 2.6 is a summary plot of the output. A plot of the output with all individual dates is shown below.


OxCal v4.4.4 Bronk Ramsey (2021); r:5 Atmospheric data from Reimer et al (2020)
Sequence FBW
Boundary Start FBW
Phase
KDE_Plot FBW
Sum Sum FBW
R_Date GrN-5103
R_Date GrA-39456
R_Date GrN-16492
R_Date GrN-16494
R_Date Poz 71757
R_Date GrN-2221
R_Date GrN-1824C
R_Date GrN-18785
R_Date GrN-12262
R_Date GrN-12263
R_Date GrN-11763
R_Date GrN-11764
R_Date GrN-11765
R_Date GrN-11766
R_Date GrN-9209
4500
4000
3500
3000
2500
Modelled date (BCE)


OxCal v4.4.4 Bronk Ramsey (2021); r:5 Atmospheric data from Reimer et al (2020)
R_Date GrM-26730
R_Date GrN-16040
R_Date GrA-41638
R_Date GrA-41639
R_Date GrA-13600
R_Date GrN-6156
R_Date GrN-4201
R_Date GrN-5767
R_Date Poz-88040
R_Date GrA-13599
R_Date GrA-13705
R_Date GrA-13958
R_Date GrA-16021
R_Date KIA-23204
R_Date KIA-23207
R_Date R-combine
R_Date GrA-13706
R_Date GrA-14086
R_Date GrA-14088
R_Date GrA-14089
4000
3800
3600
3400
3200
3000
2800
2600
2400
2200
Modelled date (BCE)


OxCal v4.4.4 Bronk Ramsey (2021); r:5 Atmospheric data from Reimer et al (2020)
R_Date GrA-14090
R_Date GrA-14095
R_Date GrA-14168
R_Date GrA-14973
R_Date GrA-15642
R_Date GrA-15459
R_Date Poz-68248
R_Date Poz-68249
R_Date Beta 421914
Boundary End FBW
Sequence CW
Boundary start CW
Phase
KDE_Plot CW
Sum Sum CW
R_Date GrN-851
R_Date GrN-1855
R_Date GrN-11918
R_Date GrA-14965
R_Date GrN-7518
4000
3500
3000
2500
2000
Modelled date (BCE)


OxCal v4.4.4 Bronk Ramsey (2021); r:5 Atmospheric data from Reimer et al (2020)
R_Date GrN-6129
R_Date GrN-6368
R_Date GrN-6349
R_Date GrA-41646
R_Date SUERC-26360
R_Date GrN-6126
R_Date GrN-8247
R_Date GrN-6724
R_Date UtC-1945
R_Date UtC-1949
R_Date GrA-56014
R_Date GrN-56013
R_Date GrA-12299
Boundary End CW
3500
3000
2500
2000
Modelled date (BCE)

4 Early and Late Flat Graves

The final Bayesian chronological model looks at the dating ranges of early flat graves (EFG) and late flat graves (LFG; see Section 2.4 in Main Text). All dates from Appendix A which are associated with cremation burials and large, complex bowls have been placed into the group LFG. Dates from Appendix A which are associated with charred wooden structures and a broader spectrum of vessels have been placed into group EFG. The two-sigma range of the start and end dates for Funnel Beaker West from Bayesian chronological model 3 (see above) are input here as probable start date of EFG and the probable end date of LFG respectively.

Model code

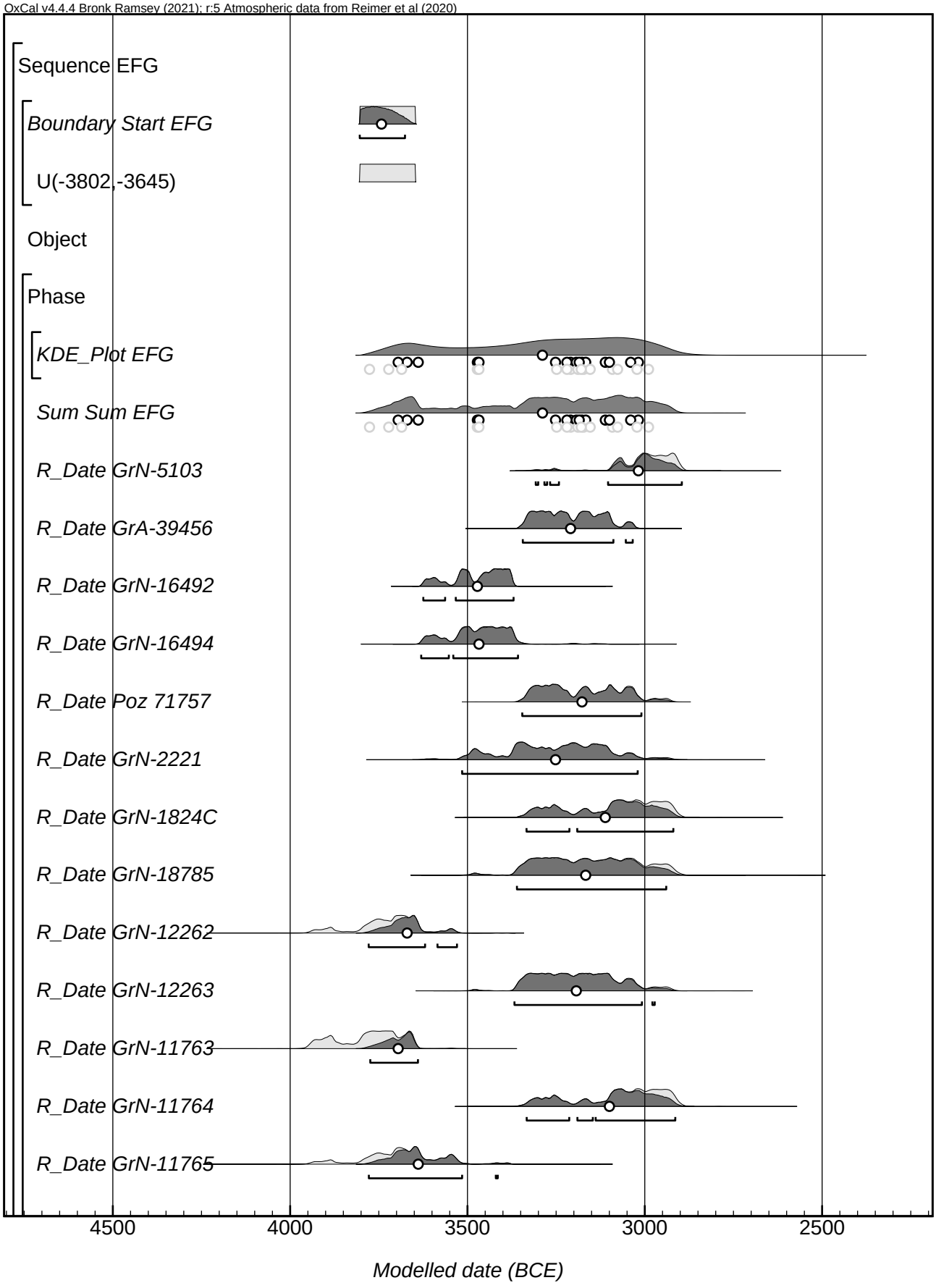
```
Plot()
 {
  Sequence("EFG")
  {
   Boundary("Start EFG",Date(U(-3802,-3645)));
   {
    color="green";
   };
   Phase()
   {
    KDE_Plot("EFG")
    {
    };
    Sum("Sum EFG")
    {
    };
    R_Date("GrN-5103",4355,45);
    R_Date("GrA-39456",4485,30);
    R_Date("GrN-16492",4700,40);
    R_Date("GrN-16494",4680,60);
    R_Date("Poz 71757",4460,40);
    R_Date("GrN-2221",4550,80);
    R_Date("GrN-1824C",4410,60);
    R_Date("GrN-18785",4460,80);
    R_Date("GrN-12262",4920,70);
    R_Date("GrN-12263",4490,60);
    R_Date("GrN-11763",4980,60);
    R_Date("GrN-11764",4400,60);
    R_Date("GrN-11765",4890,80);
    R_Date("GrN-11766",4480,60);
    R_Date("GrN-9209",4520,35);
```

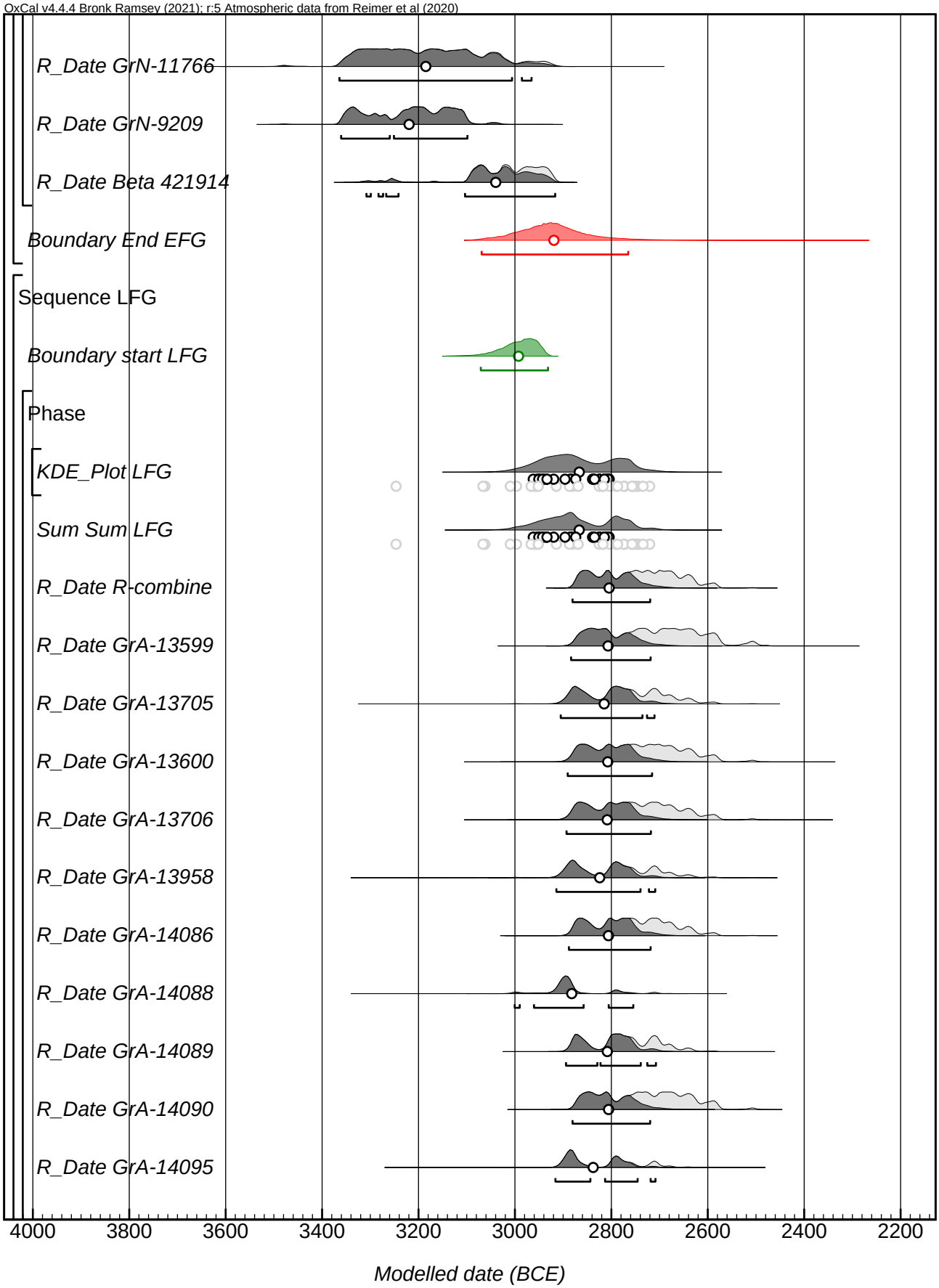
```
 R_Date("Beta 421914",4400,30);
 };
 Boundary("End EFG")
 {
 color="red";
 };
};
Sequence("LFG")
{
 Boundary("start LFG")
 {
 color="green";
 };
 Phase()
 {
 KDE_Plot("LFG")
 {
 };
 Sum("Sum LFG")
 {
 };
 R_Date("R-combine",4150,35);
 R_Date("GrA-13599",4130,50);
 R_Date("GrA-13705",4200,50);
 R_Date("GrA-13600",4160,50);
 R_Date("GrA-13706",4170,50);
 R_Date("GrA-13958",4220,50);
 R_Date("GrA-14086",4170,40);
 R_Date("GrA-14088",4270,40);
 R_Date("GrA-14089",4190,35);
 R_Date("GrA-14090",4140,40);
 R_Date("GrA-14095",4235,40);
 R_Date("GrA-14168",4170,40);
 R_Date("GrA-14973",4275,45);
 R_Date("GrA-15642",4390,60);
 R_Date("GrA-15459",4290,60);
 R_Date("GrA-16021",4230,30);
 R_Date("GrM-26730",4387,27);
 R_Date("GrN-16040",4550,60);
 R_Date("Poz-68248",4330,35);
 R_Date("Poz-68249",4220,35);
 R_Date("Poz-88040",4205,35);
 R_Date("GrA-41638",4260,35);
 R_Date("GrA-41639",4230,35);
 R_Date("GrN-4201",4380,75);
 R_Date("GrN-5767",4315,60);
 R_Date("GrN-6156",4380,40);
```

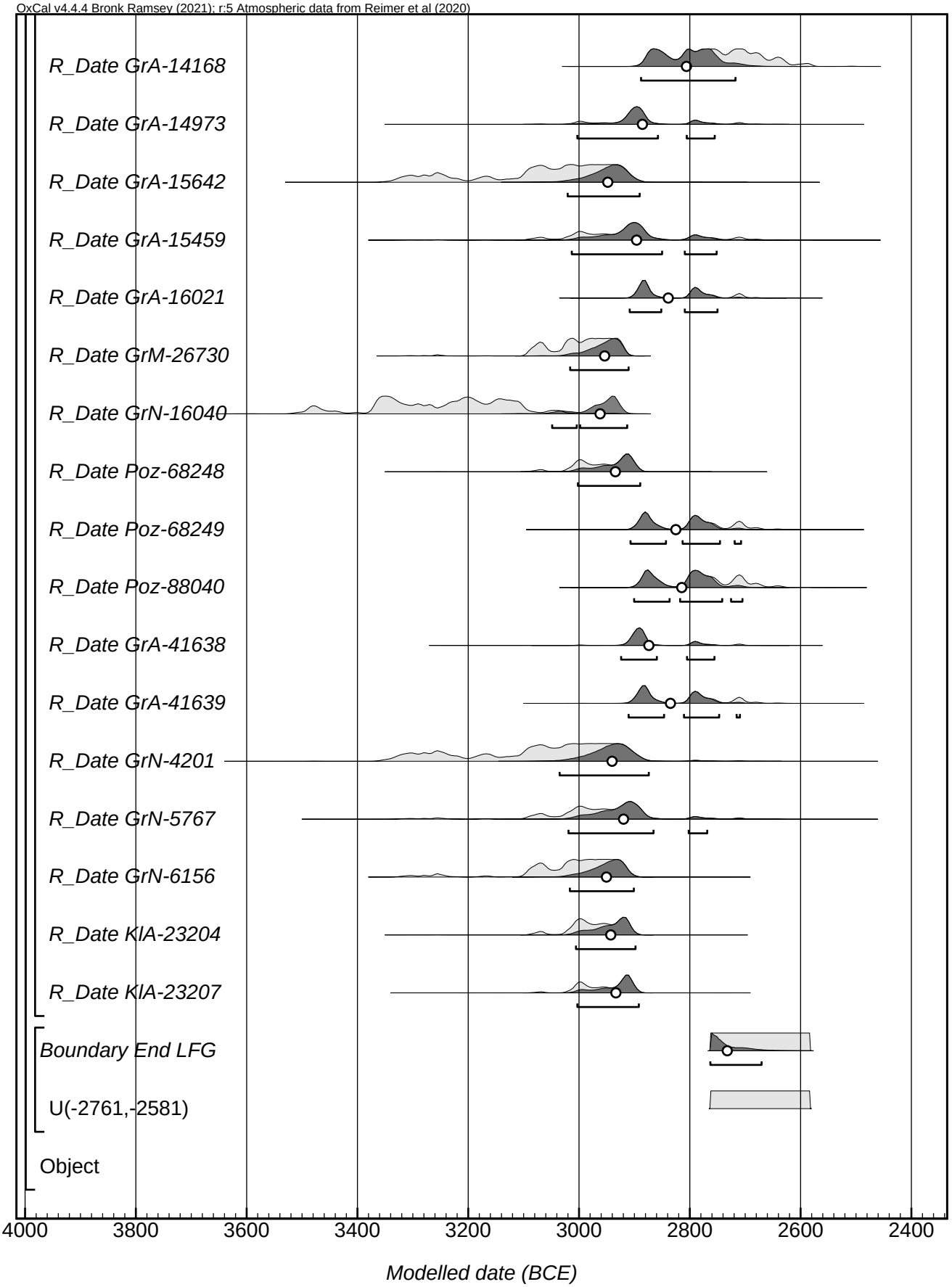
```
  R_Date("KIA-23204",4348,30);
  R_Date("KIA-23207",4331,29);
 };
 Boundary("End LFG", Date(U(-2761,-2581)));
 {
  color="red";
 };
};
};
```

Model output

The outcomes of the analysis have been attached below. The model has good fit: A_{model}=95.1, as do the individual dates $A_{overall}$=98.4. The only exception is GrN-16040 (4550 ±60 BP; 3500-3029 cal BCE; A=5.9%). Figure 2.7 in the Main Text is a summary plot of the output. A plot of the output with all individual dates is shown below.


Sequence EFG
Boundary Start EFG
U(-3802,-3645)
Object
Phase
KDE_Plot EFG
Sum Sum EFG
R_Date GrN-5103
R_Date GrA-39456
R_Date GrN-16492
R_Date GrN-16494
R_Date Poz 71757
R_Date GrN-2221
R_Date GrN-1824C
R_Date GrN-18785
R_Date GrN-12262
R_Date GrN-12263
R_Date GrN-11763
R_Date GrN-11764
R_Date GrN-11765
4500
4000
3500
3000
2500
Modelled date (BCE)


R_Date GrN-11766
R_Date GrN-9209
R_Date Beta 421914
Boundary End EFG
Sequence LFG
Boundary start LFG
Phase
KDE_Plot LFG
Sum Sum LFG
R_Date R-combine
R_Date GrA-13599
R_Date GrA-13705
R_Date GrA-13600
R_Date GrA-13706
R_Date GrA-13958
R_Date GrA-14086
R_Date GrA-14088
R_Date GrA-14089
R_Date GrA-14090
R_Date GrA-14095
4000
3800
3600
3400
3200
3000
2800
2600
2400
2200
Modelled date (BCE)


R_Date GrA-14168
R_Date GrA-14973
R_Date GrA-15642
R_Date GrA-15459
R_Date GrA-16021
R_Date GrM-26730
R_Date GrN-16040
R_Date Poz-68248
R_Date Poz-68249
R_Date Poz-88040
R_Date GrA-41638
R_Date GrA-41639
R_Date GrN-4201
R_Date GrN-5767
R_Date GrN-6156
R_Date KIA-23204
R_Date KIA-23207
Boundary End LFG
U(-2761,-2581)
Object
4000
3800
3600
3400
3200
3000
2800
2600
2400
Modelled date (BCE)